

CLASSIFICATION OF AMINO ACIDS

1. All proteins contain the

- a) Same 20 amino acids
- b) Different amino acids
- c) 300 Amino acids occurring in nature
- d) Only a few amino acids

2. Identify the amino acids containing nonpolar, aliphatic R groups.

- a) Phenylalanine, tyrosine, and tryptophan
- b) Glycine, alanine, leucine
- c) Lysine, arginine, histidine
- d) Serine, threonine, cysteine

3. The optically inactive amino acid is

- a) Glycine
- b) Serine
- c) Threonine
- d) Valine

4. At neutral pH, a mixture of amino acids in solution would be predominantly:

- a) Dipolar ions
- b) Nonpolar molecules
- c) Positive and monovalent
- d) Hydrophobic

5. Sulphur containing amino acid is

- a) Methionine
- b) Leucine
- c) Valine
- d) Asparagine

6. Selenocysteine is coded by:

- a) UAG
- b) UGA
- c) UAA
- d) GUA

7. All of the following are essential amino acids

- a) Methionine
- b) Lysine
- c) Alanine
- d) Leucine

8. Biuret reaction is specific for

- a) $-\text{CONH}-$ linkages
- b) $-\text{CSNH}_2$ group
- c) $-(\text{NH})\text{NH}_2$ group
- d) All of these

9. Which of the following is a tripeptide?

- a) Anserine
- b) Oxytocin
- c) Glutathione
- d) Kallidin

10. Which bond is present in the primary structure of protein?

- a) Ester
- b) Hydrogen
- c) Ionic bond
- d) Peptide

11. Which of the following is not an aromatic amino acid?

- a) Phenylalanine
- b) Tyrosine
- c) Tryptophan
- d) Valine

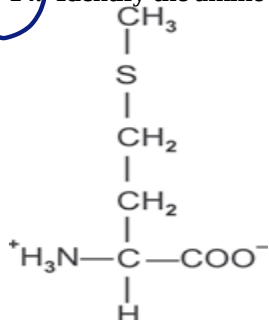
12. Which of the following group contains only nonessential amino acid?

- a) Acidic amino acid
- b) Basic amino acid
- c) Aromatic amino acid
- d) Branched chain amino acid

13. Semiessential amino acids are:

- a) Arginine
- b) Methionine
- c) Glycine
- d) Lysine

14. Identify the amino acid given in the diagram.



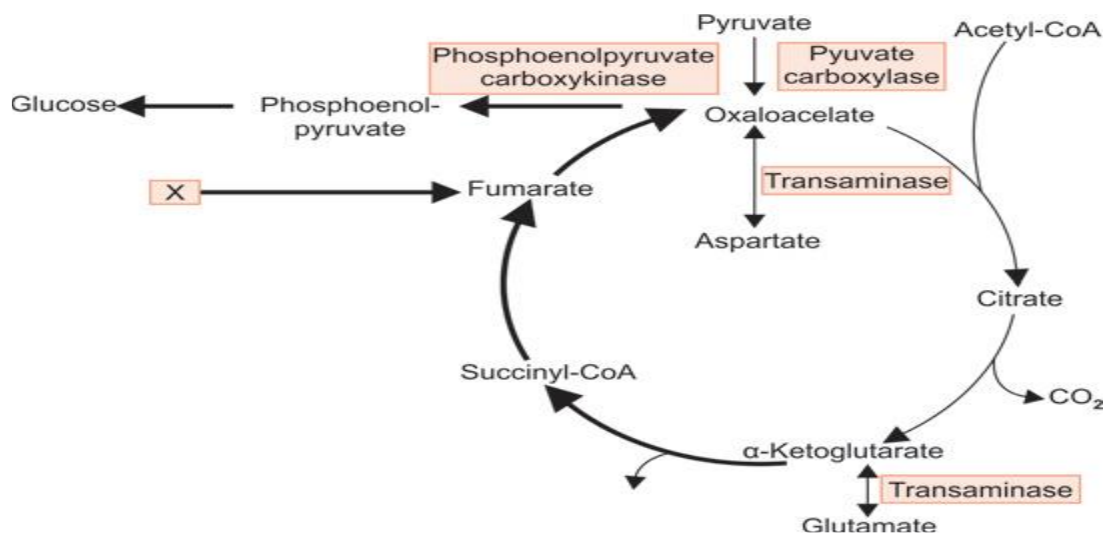
a) Cysteine

c) Methionine

b) Cystine

d) Homocysteine

15. Which of the following is the amino acid marked X, that enter at the level of Fumarate?



a) Alanine

c) Tyrosine

b) Tryptophan

d) Histidine

16. All amino acids found in proteins are optically active, except:

a) Serine

b) Glycine

c) Threonine

d) Tyrosine

17. All of the following forces may play a role in the formation of tertiary structure, except:

a) Hydrogen bond

c) Hydrophobic interaction

b) Disulfide bridges

d) Peptide bonds

18. The greatest buffering capacity at physiological pH would be provided by a protein, rich in which of the following amino acids:

a) Glycine

c) Histidine

b) Lysine

d) Valine

19. Which of the following amino acid is exclusively ketogenic?

a) Leucine

c) Threonine

b) Phenylalanine

d) Isoleucine

20. Which class of amino acids contain only nonessential amino acids?

a) Aromatic

c) Branched chain

b) Acidic

d) Basic

PROPERTIES OF AMINO ACIDS:

21. Replacing alanine by which amino acid will increase UV absorbance of protein at 280 nm wavelength:

- a) Leucine
- c) Arginine
- b) Proline
- d) Tryptophan

22. Which of the following proteins cannot be phosphorylated using protein kinase in prokaryotic organisms?

- a) Threonine
- b) Tyrosine
- c) Serine
- d) Asparagine

23. Carboxylation of clotting factors by vitamin K is required to be biologically active. Which of the following amino acid is carboxylated?

- a) Histidine
- c) Glutamate
- b) Histamine
- d) Aspartate

24. The property of proteins to absorb ultraviolet rays of light is due to:

- a) Peptide bond
- b) Imino group
- c) Disulfide bond
- d) Aromatic amino acid

25. All biologically active amino acids are:

- a) L-forms
- b) D-forms
- c) Mostly D-forms
- d) D- and L-forms

26. Optically inactive amino acid is:

- a) Proline
- b) Glycine
- c) Lycine
- d) Leucine

27. 2-Amino 3-OH propanoic acid is

- a) Glycine
- b) Alanine
- c) Valine
- d) Serine

28. Amino acids excepting histidine are not good buffering agents in cell because

- a) They exist as zwitter ions
- b) Their pK and not in the physiological pH of acell
- c) Only Histidine has pK of its R group at 6.0 unlike the others which have at a different pH
- d) None of these

29. Isoelectric pH of an amino acid is that pH at which it has a

- a) Positive charge
- b) Negative charge
- c) No net charge
- d) All of these

30. The factor which does not affect pK_a value of an amino acid is _____

- a) The loss of charge in the α -carboxyl and α -amino groups
- b) The interactions with other peptide R groups
- c) Other environmental factors
- d) Molecular weight

31. Which of the following is not the classified form of conjugated proteins?

- a) Lipoproteins
- b) Glycoproteins
- c) Metalloproteins
- d) Complete proteins

32. Which part of the amino acid gives it uniqueness?

- a) Amino group
- b) Carboxyl group
- c) Side chain
- d) None of the mentioned

33. Which of the following information is responsible to specify the three-dimensional shape of a protein?

- a) The protein's peptide bond

- b) The protein's amino acid sequence
- c) The protein's interaction with other polypeptides
- d) The protein's interaction with molecular chaperons

34. What are the following is not a factor responsible for the denaturation of proteins?

- a) pH change
- b) Organic solvents
- c) Heat
- d) Charge

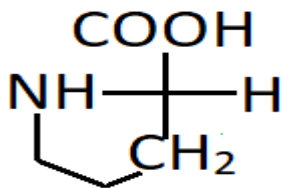
35. If the basic formula of an α -amino acid is $R-CH(NH_2)-COOH$, where R is the side chain, what is the primary point of distinction between any two proteins?

- a) Number of amino groups
- b) Number of carboxyl groups
- c) The side chain R
- d) Relative positions of amino, carboxyl groups and R

36. Cysteine is a/an _____ amino acid.

- a) Acidic
- b) Essential
- c) Aromatic
- d) Sulphur containing

37. The structure shown below is _____



- a) Side chain of histidine
- b) Side chain of tryptophan
- c) Side chain of proline
- d) Proline

38. Identify the amino acid with the formula $HOOC-CH_2-CH_2-CH(NH_2)-COOH$

- a) Glutamic acid
- b) Aspartic acid
- c) Glutamine
- d) Asparagine

39. An example of chromoprotein is

- a) Hemoglobin
- b) Sturine
- c) Nuclein
- d) Gliadin

40. The two amino acids having R groups with a negative net charge at pH 7.0 are _____

- a) Aspartate and glutamate
- b) Arginine and histidine
- c) Cysteine and methionine
- d) Proline and valine

41. Which among the following is a non-essential amino acid?

- a) Serine
- b) Threonine
- c) Lysine
- d) Histidine

42. An amino acid that yields aceto-acetyl CoA during the catabolism of its carbon skeleton will be considered as _____

- a) Glycogenic
- b) Ketogenic
- c) Both glycogenic and ketogenic
- d) Essential

43. Amino acids are mostly synthesised from

- a) Fatty acids
- b) Mineral salts
- c) α -ketoglutaric acid
- d) Volatile acids

44. Which of the following amino acids has to be supplemented in the diet?

- a) Phenylalanine
- b) Cysteine
- c) Glutamine
- d) Asparagine

45. Amino acids with aromatic side chain are

- a) Tryptophan, asparagine, tyrosine
- b) Tryptophan, threonine, tyrosine
- c) Phenylalanine, tryptophan, serine
- d) Phenylalanine, tryptophan, tyrosine

46. The first amino acid of any polypeptide chain in eukaryotes is
 a) Valine
 b) Methionine
 c) Glycine
 d) Alanine
47. Which of the following amino acids is not necessary to be taken in the diet?
 a) Histidine
 b) Threonine
 c) Serine
 d) Lysine
48. The naturally occurring form of amino acid in proteins
 a) L-amino acids only
 b) D-amino acids only
 c) Both L and D amino acids
 d) None of these
49. Sulphur containing amino acids are
 a) Cysteine and methionine
 b) Methionine and threonine
 c) Cysteine and threonine
 d) Cysteine and serine
50. Of the 20 standard amino acids, only _____ is not optically active. The reason is that its side chain _____.
 a) Alanine: is simple methyl group
 b) Glycine: is a hydrogen atom
 c) Glycine: is unbranched
 d) Lysine: contains only nitrogen
51. The element found in all amino acids is not found in carbohydrates is
 a) Sulphur
 b) Nitrogen
 c) Carbon
 d) Hydrogen
52. When the amino acid alanine (R-group is CH_3) is added to a solution with a pH of 7.3, alanine becomes
 a) A cation
 b) Nonpolar
 c) A zwitterion
 d) An isotope
53. Which of the following amino acids can form hydrogen bonds with their side (R) groups?
 a) Asparagine
 b) Aspartic acid
 c) Glutamine
 d) All of these
54. The isoelectric point of an amino acid is defined as the pH
 a) Where the molecule carries no electric charge
 b) Where the carboxyl group is uncharged
 c) Where the amino group is uncharged
 d) Of maximum electrolytic mobility
55. An essential amino acid is one that
 a) Is essentially easy to synthesize
 b) Is essential to flagella motion
 c) The body cannot synthesize
 d) The body can synthesize under essential conditions
56. In a polypeptide average mass of an amino acid residue is
 a) 110 daltons
 b) 118 daltons
 c) 80 daltons
 d) 150 daltons
57. Which pair of amino acids absorbs the most UV light at 280 nm?
 a) Threonine & Histidine
 b) Trp & Tyrosine
 c) Cysteine & Asparagine
 d) Phenylalanine & Proline
58. Albumin is deficient in which of the following amino acid?
 a) Glycine
 b) Tryptophan
 c) Cysteine
 d) Methionine
59. Which of the following is a kind of secondary structure?
 a) α -helix
 b) β -bend
 c) Triple helix
 d) All of the above
60. Which of the following is a non-protein amino acid?
 a) Proline
 b) Histidine
 c) Ornithine
 d) Asparagine

PEPTIDES AND PROTEINS

61. Proteins are formed primarily from _____ bonds.

- a) Glycosidic
- b) Peptide
- c) Phosphodiester
- d) Disulphide

62. Alanyl glycyl phenylalanine is an example of a _____

- a) Dipeptide
- b) Tripeptide
- c) Tetrapeptide
- d) Polypeptide

63. Which of the following bonds is not found in fibrous proteins?

- a) Phosphodiester
- b) Peptide
- c) Hydrogen bonds
- d) Disulphide

64. The sequence in which amino acids are arranged in a protein is called _____ structure.

- a) Primary
- b) Secondary
- c) Fibrous
- d) Sheet

65. Which type of bonds govern the secondary structure of proteins?

- a) Covalent
- b) Hydrogen
- c) Electrostatic
- d) Peptide

66. Which of the following is soluble in water?

- a) Insulin
- b) Elastin
- c) Fibroin
- d) Collagen

66. The structure in which all peptide chains are stretched out to full extension and laid side by side through intermolecular hydrogen bonds is called _____

- a) α -helix
- b) β -pleated sheet
- c) Tertiary structure
- d) Quaternary structure

67. Which amino acid when exposed to ninhydrin produces a brown color?

- a) Asparagine
- b) Proline
- c) Alanine
- d) Valine

68. Glutathione is found in all mammalian cells except:

- a) RBC
- b) Neurons
- c) Skeletal muscle
- d) Argentaffin cells

69. Plasma albumin performs the following function, except:

- a) Maintenance of osmotic pressure
- b) Transport
- c) Solubilization of glucose
- d) Nutritive

CLASSIFICATION OF ENZYMES, ENZYME KINETICS

1. Enzymes increase reaction rates by:
 - a) Selectively enhancing the rate of the forward reaction
 - b) Altering the change in free energy of the reaction
 - c) Changing the equilibrium constant of the reaction
 - d) Decreasing the rate of activation
2. The Michaelis constant (K_m) is:
 - a) Not changed by the presence of non competitive inhibitor
 - b) Equal to $1/2 V_{max}$
 - c) The substrate concentration at $1/2 V_{max}$
 - d) The equilibrium constant for the dissociation of ES to E + P
3. All are true about oxygenases, except:
 - a) Can incorporate 2 atoms of O_2 in a substance
 - b) Can incorporate 1 atom of O_2 in a substance
 - c) Important in hydroxylation of steroids
 - d) Catalyse carboxylation of drugs
4. All of the following enzymes are involved in oxidation-reduction, except:
 - a) Dehydrogenases
 - b) Hydrolases
 - c) Oxygenases
 - d) Peroxidases
5. Enzyme which cleave C-C bond:
 - a) Lyase
 - b) Oxidoreductase
 - c) Ligase
 - d) Isomerase
6. Velocity at K_m is:
 - a) Half the substrate concentration
 - b) Same as V_{max}
 - c) Quarter the V_{max}
 - d) Half the V_{max}
7. Coenzyme in decarboxylation reaction:
 - a) Niacin
 - b) Biotin
 - c) Pyridoxine
 - d) Riboflavin
8. The type of enzyme inhibition in which Succinate dehydrogenase reaction is inhibited by malonate is an example of:
 - a) Noncompetitive
 - b) Uncompetitive
 - c) Competitive
 - d) Allosteric
9. K_m changes and V_{max} remains the same. What is the type of Enzyme inhibition?
 - a) Competitive Inhibition
 - b) Noncompetitive Inhibition
 - c) Uncompetitive inhibition
 - d) Suicide Inhibition
10. All of the covalent modification regulate enzyme kinetics except:
 - a) Phosphorylation
 - b) Acetylation
 - c) ADP Ribosylation
 - d) Glycosylation
11. The following affect enzyme activity except:
 - a) Methylation
 - b) Acetylation
 - c) Induction
 - d) Phosphorylation
12. Marker enzyme for Golgi apparatus:
 - a) Galactosyltransferase
 - b) Glucose 6 Phosphatase
 - c) 5' Nucleotidase
 - d) Catalase
13. True about isoenzymes is:
 - a) Catalyse the same reaction
 - b) Same quaternary structure
 - c) Same distribution in different organs
 - d) Same enzyme classification with same number and name
14. Which of the following estimates blood creatinine level most accurately:
 - a) Jaffe method
 - b) Kinetic Jaffe method
 - c) Technicon method
 - d) Enzyme assay

15. LDH-5 level elevated in which cell injury:

- a) Liver
- b) Heart
- c) Muscle
- d) RBC

16. Which of the following LDH is having fastest electrophoretic mobility?

- a) LDH-1
- b) LDH-2
- c) LDH-3
- d) LDH-5

17. Example of an extracellular enzyme is

- a) Lactate dehydrogenase
- b) Cytochrome oxidase
- c) Pancreatic lipase
- d) Hexokinase

18. An example of ligases is

- a) Succinate thiokinase
- b) Alanine racemase
- c) Fumarase
- d) Aldolase

19. In reversible non-competitive enzyme activity inhibition

- a) V_{max} is increased
- b) K_m is increased
- c) K_m is decreased
- d) Concentration of active enzyme is reduced

20. The pH optima of most of the enzymes is

- a) Between 2 and 4
- b) Between 5 and 9
- c) Between 8 and 12
- d) Above 12

21. Hexokinase

- a) Heat stable, dialyzable, non protein organic molecules
- b) Soluble, colloidal, protein molecules
- c) Structural analogue of enzymes
- d) Different forms of enzymes

22. An example of group transferring coenzyme is

- a) NAD⁺
- b) NADP⁺
- c) FAD
- d) CoA

23. The normal serum acid phosphatase activity ranges from

- a) 5.0–13.0 KA units/100 ml
- b) 1.0–5.0 KA units/100 ml
- c) 13.0–18.0 KA units/100 ml
- d) 0.2–0.8 KA units/100 ml

24. Isoenzymes can be characterized by

- a) Proteins lacking enzymatic activity that are necessary for the activation of enzymes
- b) Proteolytic enzymes activated by hydrolysis
- c) Enzymes with identical primary structure
- d) Similar enzymes that catalyze different reaction

None of the options are correct, verify yourself

25. The isoenzymes of LDH

- a) Differ only in a single amino acid
- b) Differ in catalytic activity
- c) Exist in 5 forms depending on M and H monomer contents
- d) Occur as monomers

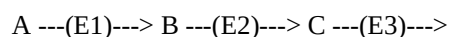
26. Which enzyme is responsible for the oxidation reaction that occurs in some foods, mostly fruits and vegetables, that turns the food brown?

- a) Polyphenol oxidase
- b) Bromelain
- c) Serrapeptidase
- d) Papain

27. Enzyme-driven metabolic pathways can be made more efficient by

- a) Concentrating enzymes within specific cellular compartments
- b) Grouping enzymes into free-floating, multienzyme complexes
- c) Fixing enzymes into membranes so that they are adjacent to each other
- d) All of the above

28. In the enzyme-catalyzed reaction shown below, what will be the effect on substances A, B, C, and D of inactivating the enzyme labeled E2?



- a) A, B, C, and D will all still be produced
c) A and B will still be produced, but not C or D
- b) A, B, and C will still be produced, but not D
d) A will still be produced, but not B, C, or D
- 29. The role of Asp 102 and His 57 during trypsin catalysis is to**
a) Neutralize the charge on the other's side chain
b) Keep the specificity pocket open
c) Function as a proton shuttle
d) Clamp the substrate into the active site
- 30. A competitive inhibitor of an enzyme is usually**
a) A highly reactive compound
b) a metal ion such as Hg^{2+} or Pb^{2+}
c) Structurally similar to the substrate
d) Water insoluble
- 31. Linear inhibition is sometimes called as**
a) complete inhibition
b) incomplete inhibition
c) partial inhibition
d) mixed inhibition
- 32. The types of inhibition pattern based on Michaelis Menten equation are**
a) competitive
b) non-competitive
c) uncompetitive
d) all of the above
- 33. The effect of non-competitive inhibition on a Lineweaver-Burk Plot is that**
a) it can move the entire curve to the right
b) it can change the y-intercept
c) it can change the x-intercept
d) all of these
- 34. The rate-determining step of Michaelis Menten kinetics is**
a) the complex formation step
b) the complex dissociation step to produce product
c) the product formation step
d) Both (a) and (c)
- 35. Which category of enzymes belongs to class two in the international classification?**
a) Hydrolases
b) Ligases
c) Transferases
d) Isomerase
- 36. The relationship between K_{eq} , K_m and V_{max} is known as**
a) Haldane equation
b) Michaelis Menten equation
c) Numerical solution approach
d) Gibbs-Helmholtz equation
- 37. The reciprocal equation for non competitive inhibition can be arranged to the equation for the**
a) Dixon plot
b) Woolf-Augustinsson-Hofstee plot
c) Eadie-Scatchard plot
d) Hanes-Woolf plot
- 38. Which of the following statements is true for enzymatically catalyzed reaction?**
a) The activation energy of the reaction is lowered so that a larger proportion of the substrate qualifies to overcome it
b) Additional substrate molecules are energized to overcome the activation energy of the reaction
c) The activation energy of the reaction is increased, thus decreasing the likelihood that any substrate molecules will overcome it
d) The activation energy of the reaction is lowered so that a fewer substrate molecules can overcome it
- 39. Which of the following common drugs is not a specific enzyme inhibitor?**
a) Iodine
b) Methotrexate
c) Sulfbnilamide
d) Penicillin
- 40. Which of these enzymes contains a Zinc (Zn) ion?**
a) Carboxypeptidase A
b) Phosphorylase B kinase
c) Tyrosine hydroxylase
d) Phosphodiesterase
- 41. A noncompetitive inhibitor of an enzyme-catalyzed reaction**
a) increases K_M and increases V_{max}
b) increases K_M and reduces V_{max}
c) reduces K_M and increases V_{max}
d) reduces K_M and reduces V_{max}

42. An allosteric inhibitor of an enzyme usually

- a) participates in feedback regulation
- b) denatures the enzyme
- c) is a hydrophobic compound
- d) causes the enzyme to work faster

43. A classical uncompetitive inhibitor is a compound that binds

- a) reversibly to the enzyme substrate complex yielding an inactive ESI complex
- b) irreversibly to the enzyme substrate complex yielding an inactive ESI complex
- c) reversibly to the enzyme substrate complex yielding an active ESI complex
- d) irreversibly to the enzyme substrate complex yielding an active ESI complex

44. The membrane proteins that catalyze active transport reactions differ from soluble enzymes as:

- a) they are permanently changed during the reaction
- b) the substrates of the reaction are all outside the cell
- c) they do not enhance the rates of reaction
- d) the products of the reaction move in a specific direction

45. Hexokinase activity in glycolysis is inhibited by

- a) glucose-6-phosphate
- b) fructose-6-phosphate
- c) fructose-1,6-biphosphate
- d) phosphofructokinase

46. Which of the following non-proteins can act as an enzyme?

- a) DNA
- b) RNA
- c) Phospholipid
- d) Glycolipid

47. All of the following serum enzymes are elevated in myocardial infarction, except:

- a) Alkaline phosphatase
- b) Lactate dehydrogenase
- c) Aspartate transaminase
- d) Creatine kinase

48. Enzymes may be used as the following, except:

- a) Therapeutic agents
- b) Nutrients
- c) Diagnostic agents
- d) Laboratory reagents

49. Enzyme activity in biological systems may be affected by the following:

- a) Negative modifiers
- b) Change in pH
- c) Change in enzyme concentration
- d) All of the above

50. Isoenzymes can be characterized as:

- a) Non-protein part of enzyme
- b) Enzymes with same quaternary structure
- c) Similar enzymes that catalyze different reactions
- d) Multiple forms of given enzyme that catalyze same type of reactions

51. Transaminase enzymes belong to the class:

- a) Hydrolases
- b) Transferases
- c) Oxidoreductases
- d) Isomerases

52. Liver diagnostic enzyme is:

- a) Alkaline phosphatase
- b) Acid phosphatase
- c) Alanine transaminase
- d) Amylase

53. Which of the following enzyme levels increases in obstructive jaundice?

- a) Lipase
- b) Amylase
- c) Acid phosphatase
- d) Alkaline phosphatase

54. The digestive enzymes belong to:

- a) Isomerases
- b) Transferases
- c) Hydrolases
- d) Ligases

55. The enzyme useful in the treatment of leukemia is:

- a) α -chymotrypsin
- b) Hyaluronidase
- c) Asparaginase
- d) Streptokinase

VITAMINS

1. The composition pyridine dependents dehydrogenases include:

- a) Vitamin B₂
- b) Vitamin B₁
- c) Vitamin PP
- d) Pantothenic acid and cytochromes

2. The composition flavin dependents dehydrogenases include:

- a) Inositol
- b) Ubiquinone
- c) Coenzyme A (CoA)
- d) Vitamin B₂

3. The structure of rhodopsin-retinal receptor protein includes:

- a) Riboflavin
- b) Calciferol
- c) Retinal
- d) Tocopherol

4. The structure of the FAD-dependent oxidoreductases include vitamin:

- a) B₃
- b) B₂
- c) B₅
- d) A

5. What vitamins are deposited in the animal organism:

- a) A, B₂, C, D
- b) B₁, H, P, E
- c) D, E, K, A
- d) A, E, D, H

6. S. Funk established structure of vitamin:

- a) PP
- b) Pyridoxine
- c) Retinol
- d) Thiamine

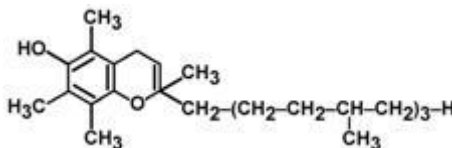
7. The sulfur atom is included in the composition of vitamins:

- a) B₆ and B₁
- b) B₁ and H
- c) H and B₃
- d) S and K

8. Vitamins are classified into:

- a) Fat-soluble, water-soluble
- b) Antivitamins
- c) Water-soluble, nonfat-vitamins
- d) Hormone

9. Indicate any vitamin formula shown below?



- a) Vitamin C
- b) Vitamin PP
- c) Vitamin E
- d) Vitamin B₁₂

10. Which substance is a member of folic acid?

- a) Ascorbic acid
- b) Paraaminobenzoic acid
- c) Pantothenic acid
- d) Isoalloxazine

11. Vitamin B₆ is involved in the reactions:

- a) Carboxylation
- b) Phosphorylation
- c) Transamination
- d) Deamination

12. Vitamin D regulates the body content:

- a) Sodium
- b) Calcium
- c) Potassium
- d) Magnesium

13. When thiamine vitamin deficiency disease develops:

- a) Rickets
- b) Anemia
- c) "Beri-Beri"
- d) Dermatitis

14. A deficiency of vitamin B₁₂ causes

- a) Beri-Beri
- b) Pernicious anemia

- c) Scurvy
- d) Iron

15. A provitamin D synthesized in human beings is

- a) Ergosterol
- b) 25-Hydroxycholecalciferol
- c) 7-Dehydrocholesterol
- d) Cholecalciferol

16. Anti-egg white injury factor is:

- a) Pyridoxine
- b) Biotin
- c) Liponic acid
- d) Thiamin

17. Deficiency of vitamin D causes

- a) Tuberculosis of bone
- b) Rickets and osteomalacia
- c) Hypothyroidism
- d) Skin cancer

18. Riboflavin deficiency symptoms are:

- a) Glossitis
- b) Stomatitis
- c) Vomiting
- d) Glossitis and Stomatitis

19. When kidney diseases are present, oral doses of vitamin D may not be effective in curing rickets, because:

- a) Hydroxylation reaction is taking place in kidney which activates vitamin
- b) Dehydrogenation of vitamin D is taking place in kidney
- c) Hydroxylation of vitamin D is taking place in kidney which destroys vitamin
- d) Vitamin D is stored in liver

20. Biochemical function of vitamin K is for:

- a) Converting proline to hydroxyproline
- b) Conversion of prothrombin to thrombin
- c) Gamma carboxylation of clotting factors
- d) Inhibition of lipid peroxidation in biomembranes

21. All are true about vitamin D metabolism, except:

- a) 1-alpha hydroxylation occurs in kidney
- b) 25-alpha hydroxylation occurs in Liver
- c) In absence of sun light, the daily requirement is 400–600 IU per day
- d) Williams syndrome is associated with mental retardation, precocious puberty and obesity

22. Vitamin A intoxication cause injury to:

- a) Lysosomes
- b) Mitochondria
- c) Endoplasmic reticulum
- d) Microtubules

23. Which of the following is true about vitamin K?

- a) Vit K dependent factors undergo post-transcriptional modification
- b) Prothrombin is a vitamin K dependent factor
- c) Stuart-Prower factor is not vitamin K dependent
- d) Menadione is a natural water insoluble vitamin K used in clinical practice

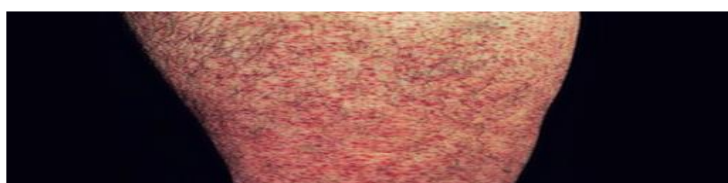
24. Pantothenate Kinase associated neuro degeneration is:

- a) Wilson's Disease
- b) Hallervorden- Spatz syndrome
- c) McLeod Syndrome
- d) Lesch-Nyhan Syndrome

25. Severe thiamine deficiency is associated with:

- a) Decreased RBC transketolase activity
- b) Increased clotting time
- c) Decreased RBC transaminase activity
- d) Increased xanthurenic acid excretion

26. Identify the vitamin deficiency given in the picture:



- a) Pellagra
- b) Scurvy

- c) Beriberi
d) Burning foot syndrome
27. The primary function of vitamin C (ascorbic acid) is to:
a) act a cofactor in enzyme catalysis
b) facilitate blood clotting
c) aid in the absorption of iron
d) promote bone mineralization
28. Which of the following vitamins is synthesized in the body from cholesterol?
a) vitamin D
b) vitamin A
c) vitamin E
d) vitamin K
29. Which of the following vitamins is required for collagen synthesis?
a) vitamin D
b) vitamin C
c) vitamin B6
d) vitamin B9
30. Which of the following is the coenzyme from of vitamin B3 (Niacin)?
a) NAD⁺
b) NADP⁺
c) coenzyme A
d) FMN
31. The main function of vitamin K is to:
a) Act as an antioxidant
b) Support immune function
c) Regulate calcium absorption
d) Facilitate blood clotting
32. Which of the following coenzymes is not derived from vitamins?
a) CoASH
b) TPP
c) Pyridoxal phosphate (PLP)
d) Coenzyme Q
33. Which of the following vitamins would most likely become deficient in a person who deve loped a completely vegetarian lifestyle?
a) Vitamin C
b) Niacin
c) Cobalamin
d) Vitamin E
34. Pyridoxal phosphate is a coenzyme for the reactions, except:
a) Transamination
b) Deamination
c) Decarboxylation
d) Oxidation-reduction
35. Both folic acid and methylcobalamin are required in:
a) Phosphorylation
b) Deamination
c) Methylation of homocysteine to methionine
d) Conversion of pyruvate to acetyl-CoA
36. Biotin is involved in:
a) Oxidation-reduction
b) Carboxylation
c) Decarboxylation
d) Dehydration
37. Both Wernicke's disease and beriberi can be treated by administering vitamin:
a) Thiamine
b) Niacin
c) Riboflavin
d) Ascorbic acid
38. All of the following vitamins have antioxidant property, except:
a) β -carotene
b) Ascorbic acid
c) Tocopherol
d) Cholecalciferol
39. Excretion of FIGLU in urine occurs in deficiency of:
a) Thiamin
b) Folic acid
c) Ascorbic acid
d) Nicotinic acid
40. Which of the following is used as an assay of riboflavin status?
a) Transketolase
b) Glutathione reductase
c) FIGLU
d) Pyruvate dehydrogenase

CARBOHYDRATE METABOLISM

1. Glucose-6-phosphatase is not present in
 - a) Liver and kidneys
 - b) Kidneys and muscles
 - c) Kidneys and adipose tissue
 - d) Muscles and adipose tissue
2. Glucose-6-phosphatase is absent or deficient in
 - a) Von Gierke's disease
 - b) Pompe's disease
 - c) Cori's disease
 - d) McArdle's disease
3. An amphibolic pathway among the following is
 - a) HMP shunt
 - b) Glycolysis
 - c) Citric acid cycle
 - d) Gluconeogenesis
4. Excessive intake of ethanol increases the ratio:
 - a) NADH : NAD⁺
 - b) NAD⁺ : NADH
 - c) FADH₂ : FAD
 - d) FAD : FADH₂
5. Which of the following enzyme catalyzes the first step of glycolysis?
 - a) Hexokinase
 - b) Pyruvate kinase
 - c) Glucokinase
 - d) Phosphofructokinase-1
6. The general test for detection of carbohydrates is
 - a) Iodine test
 - b) Molisch test
 - c) Barfoed test
 - d) Osazone test
7. Synthetase activity is depressed by
 - a) Glucose
 - b) Insulin
 - c) Cyclic AMP
 - d) Fructokinase
8. Cleavage of Fructose 1, 6-bisphosphate yields _____
 - a) Two aldoses
 - b) Two ketoses
 - c) An aldose and a ketose
 - d) Only a ketose
9. The reaction succinyl CoA to succinate requires
 - a) CDP
 - b) ADP
 - c) GDP
 - d) NADP⁺
10. The carrier of the citric acid cycle is
 - a) Succinate
 - b) Fumarate
 - c) Malate
 - d) Oxaloacetate
11. What is the first step in the payoff phase of glycolysis?
 - a) Reduction of 1, 3-bisphosphoglycerate to glyceraldehyde 3-phosphate
 - b) Oxidation of glyceraldehyde 3-phosphate to 1, 3-bisphosphoglycerate
 - c) Reversible conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate
 - d) Irreversible conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate
12. Pyruvate dehydrogenase complex and α -ketoglutarate dehydrogenase complex require the following for their oxidative decarboxylation:
 - a) CoASH and Lipoic acid
 - b) NAD⁺ and FAD
 - c) CoASH and TPP
 - d) CoASH, TPP, NAD⁺, FAD, Lipoate
13. Oxidative decarboxylation of pyruvate requires
 - a) NADP⁺
 - b) Cytochromes
 - c) pyridoxal phosphate
 - d) CoASH
14. Conversion of Alanine to carbohydrate is termed:
 - a) Glycogenesis
 - b) Gluconeogenesis
 - c) Glycogenolysis
 - d) Photosynthesis

15. The following is an enzyme required for glycolysis:

- a) Pyruvate kinase
- b) Pyruvate carboxylase
- c) Glucose-6-phosphatase
- d) Glycerokinase

16. Which substrate is used in the last step of glycolysis?

- a) Glyceraldehyde 3-phosphate
- b) Pyruvate
- c) Phosphoenolpyruvate
- d) 1, 3-bisphosphoglycerate

17. Glycogen is converted to glucose-1-phosphate by

- a) UDPG transferase
- b) Branching enzyme
- c) Phosphorylase
- d) Phosphatase

18. Which of the following is not an enzyme involved in glycolysis?

- a) Enolase
- b) Aldolase
- c) Hexokinase
- d) Glucose oxidase

19. Tissues form lactic acid from glucose. This phenomenon is termed as

- a) Aerobic glycolysis
- b) Oxidation
- c) Oxidative phosphorylation
- d) Anaerobic glycolysis

20. Fluoride inhibits _____ and arrests glycolysis.

- a) Glyceraldehyde-3-phosphate dehydrogenase
- b) Aconitase
- c) Enolase
- d) Succinate dehydrogenase

21. High concentration of glucose 6-phosphate is inhibitory to _____

- a) Hexokinase
- b) Pyruvate kinase
- c) Glucokinase
- d) Phosphofructokinase-1

22. The product formed in the first substrate level phosphorylation in glycolysis is _____

- a) Pyruvate
- b) 3-phosphoglycerate
- c) 1, 3-bisphosphoglycerate
- d) 2-phosphoglycerate

23. Glycolysis converts _____

- a) Glucose into pyruvate
- b) Glucose into phosphoenolpyruvate
- c) Fructose into pyruvate
- d) Fructose into phosphoenolpyruvate

24. Name the hormone that stimulates the process of glycolysis

- a) Growth hormone
- b) Insulin
- c) Glucagon
- d) All of the above

25. A specific inhibitor for succinate dehydrogenase is

- a) Arsenite
- b) Melouate
- c) Citrate
- d) Cyanide

26. Which of the following is not an important precursor of glucose in animals?

- a) Lactate
- b) Pyruvate
- c) Glycerol
- d) Glucose 6-phosphate

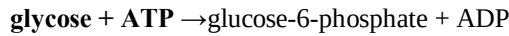
27. An essential for converting Glucose to Glycogen in Liver is

- a) Lactic acid
- b) GTP
- c) CTP
- d) UTP

28. Which of the following statements is false about gluconeogenesis?

- a) From the hydrolysis of tri-acyl-glycerol, fatty acids can be used as a carbon source
- b) From red blood cells, lactate can be used as a carbon source
- c) From the hydrolysis of tri-acyl-glycerol, glycerol is converted to glucose in gluconeogenesis
- d) From muscle vigorous muscle activity, lactate can be used as a carbon source

- 29. The carbohydrate reserved in human body is**
 a) Starch
 b) Glucose
 c) Glycogen
 d) Inulin
- 30. The reaction catalysed by phosphofructokinase**
 a) Is activated by high concentrations of ATP and citrate
 b) Uses fructose-1-phosphate as substrate
 c) Is the rate-limiting reaction of the glycolytic pathway
 d) Is inhibited by fructose 2, 6-bisphosphate
- 31. Compared to the resting state, vigorously contracting muscle shows**
 a) An increased conversion of pyruvate to lactate
 b) Decreased oxidation of pyruvate of CO₂ and water
 c) A decreased NADH/NAD⁺ ratio
 d) Decreased concentration of AMP
- 32. The concentration of blood lactate is normal:**
 a) 4,4-6,6 mmol/l
 b) 1-2 mmol/l
 c) 3,3-5,5 mmol/l
 d) 4,4-5,5 mmol/l
- 33. A basic role in digestion of carbohydrates in a digestive tract belongs following enzymes:**
 a) Alfa-amylase, hexokinase, lactase
 b) Maltase, sucrase, enterokinase
 c) Lactase, alfa-amylase, sucrase
 d) Sucrase, maltase, glycogen phosphorylase
- 34. After a well-rounded breakfast, which of the following would be expected to occur?**
 a) Increased activity of pyruvate carboxylase
 b) Decreased activity of acetyl CoA carboxylase
 c) Decreased rate of glycogenolysis
 d) Decreased rate of protein synthesis
- 35. Formation of one molecule of glucose from pyruvate requires _____**
 a) 4 ATP, 2 GTP and 2 NADH
 b) 3 ATP, 2 GTP and 2 NADH
 c) 4 ATP, 1 GTP and 2 NADH
 d) 2 ATP, 2 GTP and 2 NADH
- 36. The key regulatory enzyme of the pentose phosphate pathway is positively regulated by**
 a) Reduced nicotinamide dinucleotide (NADH)
 b) Adenosine diphosphate (ADP)
 c) Guanosine triphosphate (GTP)
 d) Nicotinamide dinucleotide phosphate (NADP⁺)
- 37. A Nigerian medical student studying in the United States develops hemolytic anemia after taking the oxidizing antimalarial drug pamaquine. This severe reaction is most likely due to:**
 a) Glucose-6-phosphate dehydrogenase deficiency
 b) Concomitant scurvy
 c) Vitamin C deficiency
 d) Diabetes
- 38. Citrate has a positive allosteric effect on which one of the following enzymes?**
 a) Pyruvate kinase
 b) Acetyl CoA carboxylase
 c) Phosphofructokinase
 d) Fatty acid synthetase
- 39. A child has ingested cyanide from her parents' garage and is rushed to the emergency room. Which of the following components of the citric acid cycle will be depleted first in this child?**
 a) NAD⁺ cofactor
 b) Citrate synthase
 c) Aconitase
 d) Citrate production
- 40. Which of the following reactions generates ATP?**
 a) Glucose-6-phosphate to fructose-6-phosphate
 b) Glucose to glucose-6-phosphate
 c) Phosphoenolpyruvate to pyruvate
 d) Pyruvate to lactate
- 41. The major metabolic product produced under normal circumstances by erythrocytes and by muscle cells during intense exercise is recycled through the liver in the Cori cycle. The metabolite is**
 a) Oxaloacetate
 b) Glycerol
 c) Lactate
- 42. Given that the standard free energy change (ΔG°) for the hydrolysis of ATP is -7.3 kcal/mol and that for the hydrolysis of glucose-6-phosphate is -3.3 kcal/mol, what is the ΔG° for the phosphorylation of glucose?**



- a) -10.6 kcal/mol
c) **-4.0 kcal/mol**
- b) -7.3 kcal/mol
d) **+4.0 kcal/mol**
- 43. Keratansulphate is found in abundance in:**
a) Heart muscle
c) Adrenal cortex
b) Liver
d) **Cornea**
- 44. Gluconeogenesis is decreased by**
a) Glucagon
c) Glucocorticoids
b) Epinephrine
d) **Insulin**
- 45. During starvation, ketone bodies are used as a fuel by**
a) Erythrocytes
c) Liver
b) **Brain**
d) All of these
- 46. Glucose is absorbed by gastrointestinal cells by:**
a) Passive diffusion
b) **Carrier mediated co- transport system, along with sodium**
c) Carrier mediated antiport system, along with potassium
d) Needs no ATP
- 47. The product of glycolysis in erythrocytes is:**
a) NADPH
b) **Lactate**
c) Pyruvate
d) Carbon dioxide
- 48. Glucokinase is more active after a meal, because:**
a) **It is an inducible enzyme**
b) It has more affinity to glucose than hexokinase
c) It is present in all tissues
d) Can act on all monosaccharides
- 49. Which enzyme catalyzes an irreversible reaction?**
a) Transketotase
b) **Phosphofructokinase**
c) aldolase
d) Glyceraldehyde-3-phosphate dehydrogenase
- 50. Complete oxidation of one molecule of glucose yields how many ATPs?**
a) 12
b) 24
c) **32**
d) 129
- 51. Catalytic activity of phosphofructokinase is increased by all the following, except:**
a) AMP
b) Fructose-2,6-bisphosphate
c) **ATP**
d) Fructose-6- phosphate
- 52. All the following coenzymes are involved in the pyruvate dehydrogenase reaction, except:**
a) Thiamine pyrophosphate (TPP)
b) **Biotin**
c) NAD^+
d) FAD
- 53. Malate shuttle is necessary for:**
a) **Glucogen**
b) Growth hormone
c) **Insulin**
d) Glucocorticoids
- 54. Blood glucose level can be raised by gluconeogenesis only by liver because:**
a) Glucokinase is present exclusively in liver
b) **Glucose-6-phosphatase is active only in liver**
c) Transaminase are present in liver
d) Liver is the site of storage of glucose as glycogen
- 55. HMP pathway is essential for all, except:**
a) Transparency of lens
b) Integrity of RBC membrane
c) Reduce methemoglobin level
d) **Provide 2,3-BPG for RBCs**

56. Which of the following drugs can precipitate GPD deficiency?

- a) Primaquan
- b) Aspirin
- c) Statin
- d) Penicilin

57. Lactate is the product of glycolysis in RBCs because:

- a) RBC can convert lactate to glucose
- b) Formation of lactate produces more energy
- c) NAD^+ is regenerated in the cytoplasm
- d) Accumulation of lactate activates glycolysis

58. The normal fasting plasma glucose level is:

- a) 40-60 mg/100 ml
- b) 70-110 mg/100 ml
- c) 120-150 mg/100 ml
- d) 60-180 mg/100 ml

59. Glucose level in plasma is increased by:

- a) Gluconeogenesis
- b) Glycolysis
- c) Glycogen synthesis
- d) Glycosuria

60. Blood is collected in fluoride oxalate bottle to:

- a) Prevent clotting
- b) Preserve glucose
- c) Preserve glucose and prevent clotting
- d) Get quick results

61. Which of the following lab results indicate poor glycemic control?

- a) PPBS of 180 mg/dl
- b) Glyconemoglobin of 10 %
- c) Serum cholesterol of 240 mg/dl
- d) Blood urea level of 35 mg/dl

62. One of the following enzymes in glycolysis catalyses an irreversible action

- a) Hexokinase
- b) Pyruvate kinase
- c) Phosphofructokinase
- d) All of them

63. The hormone that lowers c-AMP concentration in liver cells is:

- a) Glucagon
- b) Insulin
- c) Epinephrine
- d) Thyroxine

64. The connecting link between HMP shunt and lipid synthesis

- a) Ribose
- b) NADPH
- c) Sedoheptulose-7-phosphate
- d) NADH

65. Glucose and Galactose are:

- a) Anomers
- b) Constituents of sucrose
- c) Diastereoisomers
- d) Epimers

66. D-mannose is:

- a) The 4th epimer of glucose
- b) A keto sugar
- c) Is a component of maltose
- d) Is the 2nd epimer of glucose

67. A pair of sugars differing from each other in the functional group is called:

- a) Anomers
- b) epimers
- c) rasemers
- d) Stereoisomers

None of the options are correct but stereoisomers are closest correct option

68. Reducing sugars are differentiated from nonreducing sugars by:

- a) Iodine test
- b) Molish's test
- c) Selwanoff's test
- d) Benedict's test

69. In the of monosaccharides, all properties go hand in hand, except:

- a) Reducing property
- b) Formation of furfural
- c) Formation of ozonide
- d) Mutarotation

70. All the following are sugars alcohols, except:

- a) Dulcitol
- b) Mannitol
- c) Xylose
- d) Sorbitol

71. The glucosidic linkage seen in sucrose is:

- a) Alpha-1,4-linkage
- b) β -1,4-linkage
- c) alpha-1,6-linkage
- d) 1,2-linkage

72. All the following have glycosidic bond, except

- a) Maltose
- b) Sucrose
- c) N-acetylglucosamine
- d) Alpha methyl glucose

73. The sugar found in milk is:

- a) Galactose
- b) Glucose
- c) Fructose
- d) Lactose

74. The glycosidic linkage seen in maltose is:

- a) Alpha-1,4-linkage
- b) β -1,4-linkage
- c) Alpha-1,6-linkage
- d) Alpha-1,2-linkage

75. Ethanol is oxidized to acetaldehyde in the liver cytoplasm by _____

- a) Alcohol dehydrogenase
- b) Alcohol carboxylase
- c) Pyruvate carboxylase
- d) Pyruvate kinase.

76. Which of the following statements about gluconeogenesis is correct?

- a) Pyruvate is first converted to phosphoenolpyruvate by phosphoenolpyruvate carboxykinase
- b) Fructose 1, 6-bisphosphatase converts fructose 1, 6-bisphosphate into fructose 1-phosphate
- c) Glucose 6-phosphatase hydrolyzes glucose 6-phosphate to release glucose into the blood
- d) Glucose 6-phosphatase hydrolyzes glucose 6-phosphate and is found in liver and muscle

77. Which of the following enzyme is not involved in galactose metabolism?

- a) Glucokinase
- b) Galactokinase
- c) Galactose-1-Phosphate Uridyl transferase
- d) UDP-Galactose 4- epimerase

78. Which of the following enzyme is defective in galactosemia (type I) - a fatal genetic disorder in infants?

- a) Glucokinase
- b) Galactokinase
- c) Galactose-1-Phosphate Uridyl transferase
- d) UDP-Galactose 4- epimerase

79. In the liver, the accumulation of which of the following metabolite attenuates the inhibitory of ATP on phosphofructokinase?

- a) Glucose-6-Phosphate
- b) Citrate
- c) Fructose-1,6-Bisphosphate
- d) Fructose-2,6-Bisphosphate

80. Mutation in which of the following enzymes leads to a glycogen storage disease known as "Tarui's disease"?

- a) Glucokinase
- b) Phosphofructokinase
- c) Phosphoglucomutase
- d) Pyruvate Kinase

81. Erythrocytes undergo glycolysis for the production of ATP. The deficiency of enzyme leads to hemolytic anemia.

- a) Glucokinase
- b) Phosphofructokinase
- c) Phosphoglucomutase
- d) Pyruvate Kinase

82. Cancer cells have high energy demands for replication and division. Increased flux of glucose into glycolysis replenishes the energy demand. Which of the following enzyme plays an important role in tumor metabolism?

- a) Glucokinase
- b) Phosphofructokinase
- c) Phosphoglucomutase
- d) Pyruvate Kinase M2

83. Which of the following glucose transporter (GLUT) is important in insulin-dependent glucose uptake?

- a) GLUT1
- b) GLUT2
- c) GLUT3
- d) GLUT4

84. Which of the following glucose transporter (GLUT) is present in the beta cells of the pancreas?
 a) GLUT1
 b) GLUT2
 c) GLUT3
 d) GLUT4
85. Which of the following glucose transporter (GLUT) is important in fructose transport in the intestine?
 a) GLUT1
 b) GLUT3
 c) GLUT5
 d) GLUT7
86. Which of the following metabolite negatively regulates pyruvate kinase?
 a) Fructose-1,6-Bisphosphate
 b) Citrate
 c) Acetyl CoA
 d) Alanine
87. In absence of oxygen, pyruvate is converted into lactate in muscle because
 a) Lactate is the substrate from the downstream pathway
 b) Lactate acts as a substrate for the formation of amino acid
 c) During the product of lactate two ATP are produced
 d) During lactate formation, NADH is reconverted into NAD.
88. Which of the following glycolytic enzyme is inhibited by the accumulation of long-chain fatty acid in the liver?
 a) Hexokinase
 b) Glucokinase
 c) Phosphofructokinase
 d) Pyruvate kinase
89. Which of the following statement about Phosphofructokinase-2 (PFK-2) is false?
 a) PFK-2 is a bifunctional enzyme having a kinase domain, phosphatase domain, and a regulatory domain
 b) Activated protein kinase A phosphorylates PFK-2 and activates the phosphatase domain
 c) PFK-2 catalyzes the conversion of fructose-6-phosphate to fructose 2-6 bisphosphate
 d) None of the option are wrong
90. Which of the following hormone helps in regulating blood sugar (glucose) levels and metabolism in the body, and promotes the uptake of glucose by cells, especially muscle, adipose (fat), and liver cells?
 a) Glucagon
 b) Epinephrine
 c) Cortisol
 d) Insulin
91. Which of the following statement is true regarding glycolysis pathway?
 a) Glycolysis occurs only in mammalian cells
 b) Glycolysis occurs in mitochondria
 c) Glycolysis occurs in the presence and absence of oxygen
 d) Glycolysis occurs when ATP concentration is high.
92. What is the rate-limiting enzyme in glycolysis?
 a) Hexokinase
 b) Phosphohexose isomerase
 c) Glyceraldehyde-3-phosphate dehydrogenase
 d) Enolase
93. The net gain of adenosine triphosphate (ATP) during the conversion of glucose to pyruvate is:
 a) 1 ATP
 b) 2 ATP
 c) 1 ATP +1 GTP
 d) 4 ATP
94. During the conversion of glucose to pyruvate, two NADH molecules are generated. Which of the following steps generates NADH?
 a) Conversion of fructose-6-phosphate to fructose-1-6-bisphosphate
 b) Conversion of glyceraldehydes-3-phosphate to 1-3-bisphosphoglycerate
 c) Conversion of 3-phosphoglycerate to 2-phosphoglycerate
 d) Conversion of phosphoenolpyruvate to pyruvate
95. What is the committed step in glycolysis?
 a) Conversion of glucose to glucose-6-phosphate
 b) Conversion of fructose-6-phosphate to fructose-1,6-bisphosphate
 c) Conversion of glyceraldehydes-3-phosphate to 1-3-bisphosphoglycerate
 d) Conversion of 3-phosphoglycerate to 2-phosphoglycerate

96. Glycolysis consists of three irreversible steps.

Which of the following enzyme-catalyzed reaction are not irreversible steps in glycolysis?

- a) Hexokinase
- b) Phosphofructokinase
- c) Glyceraldehyde-3-phosphate kinase
- d) Pyruvate kinase

97. All of the following are the negative regulators of phosphofructokinase(PFK) and can inhibit it's activity, except?

- a) Adenosine triphosphate (ATP)
- b) Adenosine monophosphate (AMP)
- c) Citrate
- d) pH

98. Which of the following enzyme is inhibited by sodium fluoride that is used as anticoagulant during blood collection and transport?

- a) Glyceraldehyde-3-phosphate dehydrogenase
- b) 3-phosphoglycerate mutase
- c) Enolase
- d) AMP

99. During gluconeogenesis, the three irreversible steps of glycolysis have to be bypassed. The first step is the conversion of pyruvate to phosphoenolpyruvate. Which of the following statement is false regarding the reaction step?

- a) This reaction involves a two-step process catalyzed by pyruvate carboxylase and phosphoenolpyruvate carboxykinase
- b) Conversion of oxaloacetate from pyruvate occurs in mitochondria and shuttled into the cytosol.
- c) Formation of phosphoenolpyruvate requires both ATP and GTP as an energy source.
- d) Acetyl CoA is an activator of the enzyme pyruvate carboxylase.

100. During gluconeogenesis, the three irreversible steps of glycolysis have to be bypassed. The final step is the conversion of glucose-6-P to glucose which is catalyzed by glucose-6-phosphatase.

Which of the following statement is true about the reaction step?

- a) Conversion of glucose-6-phosphate to glucose releases one ATP molecule
- b) It is a highly active enzyme in skeletal muscle
- c) Defect in glucose-6-phosphatase leads to abnormal accumulation of glycogen in the liver
- d) The reaction occurs in mitochondria

101. Which of the following statement is true about Cori Cycle?

- a) The Cori cycle involves three tissues muscle, liver, and brain
- b) It involves the transport of lactate from the liver to skeletal tissue for gluconeogenesis
- c) It involves the transport of lactate from skeletal muscle to the liver for gluconeogenesis
- d) It is active during resting stages and in well-fed condition

102. During prolong starvation, which of the following hormone is responsible for increasing gluconeogenesis in the liver?

- a) Insulin
- b) Glucagon
- c) TSH
- d) Thyroxine

103. During catabolism, only about 40% of the energy available from oxidizing glucose is used to synthesize ATP. Remaining 60%:

- a) is lost as heat
- b) is used to reduce NADP
- c) remains in the products of metabolism
- d) is stored as fat.

104. Why does the glycolytic pathway continue in the direction of glucose catabolism?

- a) There are essentially three irreversible reactions that act as the driving force for the pathway
- b) High levels of ATP keep the pathway going in a forward direction
- c) The enzymes of glycolysis only function in one direction
- d) Glycolysis occurs in either direction

Krebs cycle. TCA Cycle.

1. Which of the following is not an intermediate of TCA Cycle?

- a) Acetyl CoA
- b) Citrate
- c) Succinyl CoA
- d) Alpha Ketoglutarate

2. Which of the following is true about Krebs Cycle?

- a) Pyruvate condenses with Oxaloacetate to form Citrate
- b) Alpha ketoglutarate is a five Carbon compound
- c) Oxidative Phosphorylation occurs in the cytoplasm only
- d) Krebs cycle can operate in anaerobic condition

3. Which of the following substance binds to CoA and condenses oxaloacetate to inhibit the TCA cycle:

- a) Malonate
- b) Arsenite
- c) Fluoroacetate
- d) Fumarate

4. First substrate of Krebs cycle is:

- a) Pyruvate
- b) Glycine
- c) HCl
- d) Lipoprotein

5. Hyperammonemia inhibits TCA cycle by depleting:

- a) Oxaloacetate
- b) Alpha ketoglutarate
- c) Citrate
- d) Succinyl CoA

6. False about reducing equivalents is:

- a) They are NADH and NADPH
- b) Only produced during primary metabolic pathway
- c) Formed in TCA cycle
- d) Formed in mitochondria

7. All contain high energy bond, except:

- a) ATP
- b) Glucose-6-phosphate
- c) Acetyl-CoA
- d) Phosphoenolpyruvate

8. Coenzyme Q catalyzes electron transport between:

- a) FADH and cytochrome B
- b) It is the last member in the electron transport chain
- c) NADH and ubiquinone
- d) Cytochrome Q and cytochrome C

9. Death due to cyanide poisoning is a result of:

- a) Cyanide hemoglobin complex formation
- b) Cyanide inhibiting complex I of respiratory chain
- c) Cyanide inhibiting cytochrome oxidase
- d) Cyanide blocking oxygen transport in blood

10. Which is true with cytochrome reductase?

- a) It catalyzes the electron transport between FADH and cytochrome b
- b) It is the last member in the electron transport chain
- c) It catalyzes the electron transport between NADH and ubiquinone
- d) It catalyzes the electron transport between CoQ and cytochrome C

11. Why is the TCA cycle the central pathway of metabolism of the cell?

- a) It occurs in the center of the cell
- b) Its intermediates are commonly used by other metabolic reactions
- c) All other metabolic pathways depend upon it
- d) None of the above

12. In what form does the product of glycolysis enter the TCA cycle?

- a) Acetyl-CoA
- b) Pyruvate
- c) NADH
- d) Glucose

13. Malate-aspartate shuttle operates in

- a) Lungs and liver
- b) Heart and liver
- c) Pancreas and liver
- d) None of these

14. Oxidation of a molecule involves

- a) Gain of electron
- b) Loss of electron
- c) Gain of proton
- d) Loss of proton

- 15. During cellular respiration, most of the ATP made, is generated by**
 a) Oxidative phosphorylation
 b) Photophosphorylation
 c) Glycolysis
 d) Substrate-level phosphorylation
- 16. To stop ATP synthesis which chemical is generally used?**
 a) DNSA
 b) 2,4 dinitrophenol
 c) DDT
 d) None of the chemical can stop ATP synthesis
- 17. The enzymes of the TCA cycle in a eukaryotic cell are located in the**
 a) Nucleus
 b) Mitochondria
 c) Plasma membrane
 d) Lysosomal bodies
- 18. FAD is reduced to FADH₂ during**
 a) Electron transport phosphorylation
 b) Lactate fermentation
 c) Krebs cycle
 d) Glycolysis
- 19. During glycolysis, electrons removed from glucose are passed to**
 a) FAD
 b) NAD⁺
 c) Acetyl CoA
 d) Pyruvic acid
- 20. The carbon dioxide is primary a product of**
 a) Krebs cycle
 b) Glycolysis
 c) Electron transport phosphorylation
 d) Lactate fermentation
- 21. What is the pH of the medium when sucrose is used as substrate for the production of citric acid?**
 a) 3
 b) 4
 c) 5
 d) 6
- 22. All of the following cells are dependent on glycolysis except?**
 a) Neurons
 b) RBC
 c) Renal medulla
 d) Red muscle fibres
- 23. The irreversible steps of glycolysis are all except**
 a) Pyruvate kinase
 b) Phosphoglycerate kinase
 c) Hexokinase
 d) Phosphofructokinase
- 24. The steps in which oxidative phosphorylation occurs in glycolysis are:**
 a) Pyruvate kinase
 b) Phosphoglycerate kinase
 c) PFK
 d) Glyceraldehyde 3 Phosphate Dehydrogenase
- 25. In aerobic glycolysis, lactate is formed**
 a) For the generation of ATP
 b) For the regeneration of lactate
 c) For the regeneration of pyruvate
 d) For the regeneration of NAD
- 26. The significance of Rapoport Leubering Shunt is**
 a) ATP production
 b) Lactate formation
 c) Source of fatty acids
 d) 2,3 BPG production
- 27. The enzyme involved in Rapoport Leubering Shunt is**
 a) Phosphoglycerate kinase
 b) Bisphosphoglycerate formation
 c) Bisphosphoglycerate mutase
 d) Phosphoglycerate mutase

Pentose phosphate pathway

1. HMP shunt is required for which kind of metabolism?
a) carbohydrate metabolism
b) fat metabolism
c) lipid metabolism
d) amino acid metabolism
2. What is the product generated after the completion of 1st phase?
a) ribulose-5-phosphate
b) CO₂
c) NADPH
d) all of the above
3. What is the location of pentose phosphate pathway to take place?
a) cell membrane
b) cytosol
c) ribosomes
d) mitochondria
4. What type of reactions occurs in 2nd phase?
a) oxidative reversible
b) non-oxidative reversible
c) non-oxidative irreversible
d) oxidative irreversible
5. What is the role of NADPH in RBCs?
a) transports O₂
b) forms Hb
c) keeps iron of Hb in ferrous form
d) none of the above
6. Trans aldolase enzyme catalyzes the transfer of how many carbon units from sedoheptulose-7-phosphate?
a) 2 units
b) 3 units
c) 4 units
d) 5 units
7. Which of the following statement is NOT true?
a) HMP shunt stands for hexose monophosphate shunt
b) HMP shunt does not generate CO₂
c) HMP does not generate ATP
d) pentose phosphate pathway takes place in cytosol
8. What stimulates the pentose phosphate pathway?
a) high concentration of insulin
b) low level of NADPH
c) high level of NADPH
d) both A and B
9. Which enzyme requires thiamine pyrophosphate as the co-enzyme?
a) transketolase
b) ribulose-5-phosphate isomerase
c) transaldolase
d) glucose-6-phosphate dehydrogenase
10. What is the product generated after the transketolase catalyzes the transfer of 2 carbon units from xylulose-5-phosphate to erythrose-4-phosphate?
a) fructose-4-phosphate
b) glyceraldehyde 3 phosphate
c) fructose-6-phosphate
d) both B and C
11. Which enzyme belongs to the 2nd phase of HMP shunt?
a) glucose-6-phosphate dehydrogenase
b) 6-phosphogluconate
c) 6-phosphogluconate dehydrogenase
d) none of the above
12. What type of reactions occurs in the 1st phase of pentose phosphate pathway?
a) oxidative reversible
b) non-oxidative reversible
c) non-oxidative irreversible
d) oxidative irreversible
13. What is the function of ribulose-5-phosphate epimerase?
a) epimerization of glucose-6-phosphate
b) epimerization of ribulose-5-phosphate
c) epimerization of xylulose-5-phosphate
d) epimerization of fructose -4-phosphate
14. Which form of energy is used by the glucose-6-phosphate dehydrogenase enzyme?
a) ATP
b) AMP
c) GTP
d) NADPH
15. A 24-year-old woman presents with diarrhea, dysphagia, jaundice and white transverse lines on her fingernails (Mees lines). She is diagnosed with arsenic poisoning. Arsenic poisoning inhibits which of the following enzymes?
a) Citrate synthase
b) Isocitrate dehydrogenase

c) Succinate dehydrogenase

d) α -Keto glutarate dehydrogenase complex

16. A biochemistry graduate student isolated all enzymes of the TCA cycle to produce NADH. Oxidation of which of the following substrates in the citric acid cycle is not coupled to NADH production?

a) succinate

b) Malate

c) α -Ketoglutarate

d) Isocitrate

17. A 3-year-old boy presents to the pediatric clinic with hypotonia, lactic acidosis, seizures. After an extensive workup, he is diagnosed with pyruvate dehydrogenase complex deficiency. Which of the following cofactors is not required by this to convert pyruvate to acetyl-CoA?

a) Thiamine

b) Lipoic acid

c) Ascorbic acid

d) Pantothenic acid

18. Pyruvate dehydrogenase complex deficiency is an autosomal disorder and leads to metabolic acidosis. Which of the following accumulates to cause metabolic acidosis?

a) Beta-hydroxybutyric acid

b) Acetoacetic acid

c) Lactic acid

d) Hydrochloric acid

19. A 3-year-old is presented with a history of recurrent upon sun exposure and passage of purple colored urine. The child is diagnosed with congenital erythropoietic porphyria, a disorder of the pathway of heme biosynthesis. Which of the following intermediate of the TCA cycle is used as a precursor for heme biosynthesis?

a) Succinyl – CoA

b) Acetyl-CoA

c) Malate

d) Pyruvate

20. A 16-year-old male comes to the a routine sports physical examination. His physical examination is unremarkable except for the presence of multiple pale masses on the skin, identified as xanthomas. Xanthomas are generally associated with underlying hypercholesterolemia. Which of the following is utilized as a precursor for cholesterol biosynthesis?

a) Succinyl – CoA

b) Acetyl-CoA

c) Malate

d) Pyruvate

21. A 3-year-old child is brought to the pediatric emergency with convulsions. The child is diagnosed with ammonia intoxication due to a urea cycle disorder. Reduced formation of GABA is identified as the critical cause of the convulsions, stemming from glutamate depletion, as GABA is synthesized through the decarboxylation of glutamate. Which of the following TCA cycle intermediates is involved in the formation of glutamate?

a) Succinate

b) α -Ketoglutarate

c) Isocitrate

d) Malate

22. A 5-year-old child was rushed to the pediatric emergency after accidentally consuming fluoroacetate, a known inhibitor of the TCA cycle. Which of the following enzymes is inhibited by fluoroacetate?

a) citrate synthase

b) aconitase

c) Succinate dehydrogenase

d) Isocitrate dehydrogenase

23. Malonate is an inhibitor of which of the following enzymes?

a) citrate synthase

b) aconitase

c) Succinate dehydrogenase

d) Isocitrate dehydrogenase

24. At the pre-diabetic stages and in type 2 diabetes, there may be hyperinsulinemia jointly with a paradoxical increase in glucagonemia. This may be due to:

a) α cells resistance to insulin

b) physiological response trying to compensate for the lack of insulin

c) inhibition in enzymatic response to insulin action

d) decrease in activity of enzymes that degrade glucagon's second messengers

25. Which of the following compounds inhibits oxidative phosphorylation by inhibiting the transporter of ADP into ATP out of the mitochondrion?

a) Malonate

b) Oxamate

c) Atractyloside

d) Cyanide

26. The energy of oxidation is initially trapped as a high-energy phosphate compound and then used to form ATP. Which of the following intermediates of glycolysis is high energy compound?
- Fructose-6-P
 - Glyceraldehyde-3-P
 - Fructose-1,6-bisphosphate
 - Phosphoenol pyruvate
27. Aspirin in a high dosage produces hyperthermia, despite the fact that it is itself an antipyretic agent. The reason for this effect is-
- Aspirin affects hypothalamic functions
 - It uncouples oxidative phosphorylation
 - It stimulates metabolism
 - It is itself metabolized to high energy compound
28. For each acetyl-CoA oxidized by the citric cycle, what is the energy gain by substrate-level phosphorylation?
- 2 ATP
 - 4 ATP
 - 6 ATP
 - 1 ATP
29. Which of the components of the electron transport chain does not contain the Iron-sulfur center?
- NADH dehydrogenase complex
 - Cytochrome a-a3 oxidase
 - Succinate dehydrogenase
 - Cytochrome bc₁-c reductase
30. In biological oxidation process, depending on the organic load, the oxidation takes place _____
- 1-4 Hours
 - 4-8 Hours
 - 8-16 Hours
 - 16-20 Hours
31. All impurities are oxidised into the _____ and _____ in the biological oxidation process.
- Oxygen and water
 - Carbondioxide and water
 - Nitrogen and water
 - Chlorine and water
32. What products of glucose oxidation are essential for oxidative phosphorylation?
- pyruvate
 - NADPH and ATP
 - Acetyl-CoA
 - NADH and FADH₂
33. Which of the following enzymes takes part in the malate shuttle?
- Aspartate transaminase
 - Alkaline phosphatase
 - Aconitase
 - Citrate synthase
34. Cyanide is a poison that inhibits the electron transport chain by creating a strong and stable bond with Fe-Cu center in cytochrome C oxidase (complex IV). What is the immediate consequence of cyanide poisoning?
- Prevent reduction of NADH
 - Prevent oxidation of oxygen
 - Prevent oxidation of NADH
 - Prevent reduction of oxygen
35. Which electron carrier would have the greatest negative impact on ATP production during oxidative phosphorylation if its production was inhibited?
- NADH
 - Water
 - Oxygen
 - FADH₂
36. What would occur if all available hydrogen ions were used by ATP synthase?
- It would increase the pH of the mitochondrial intermembrane space
 - It would increase the levels of inorganic phosphate in the mitochondrial matrix
 - It would increase the pH of the mitochondrial matrix
 - It would decrease the levels of inorganic phosphate in the intermembrane space
37. Which complex of the electron transport chain in cellular respiration does not directly impact the intermembrane space's pH?
- Complex II
 - Complex I
 - Complex IV
 - Complex III
38. The proton pump in oxidative phosphorylation creates a gradient of protons across the inner mitochondrial membrane. What kind of energy is generated with such a gradient?
- Membrane energy
 - Kinetic energy
 - Potential energy
 - Protonic energy

39. Disruption of which process will have the greatest impact on the number of electron carriers used by the electron transport chain?
- a) Anaerobic pathways
 - b) Formation of FADH_2
 - c) Glycolysis
 - d) The citric acid cycle
40. What is the primary function of purine and pyrimidine metabolism in the body?
- a) Energy production
 - b) Nucleotide synthesis
 - c) Hormone production
 - d) Amino acid synthesis

Hormones

1. Binding of the following hormones to receptor activate adenylate cyclase, except:
 - a) Vasopressin
 - b) Glucagon
 - c) Thyroxine
 - d) Epinephrine
2. The following are the intracellular second messenger for hormone action, except:
 - a) c-AMP
 - b) c-GMP
 - c) Phosphatidylinositol
 - d) NADPH
3. Water soluble hormones have cell surface receptors, except:
 - a) Glucagon
 - b) Epinephrine
 - c) Thyroxine
 - d) Vasopressin
4. Hormone that have intracellular receptor:
 - a) Glucocorticoids
 - b) ACTH
 - c) TSH
 - d) Glucagon
5. Second messenger that mobilizes intracellular calcium is:
 - a) Phosphatidic acid
 - b) Diacylglycerol
 - c) Inositol triphosphate
 - d) c-GMP
6. A dietary deficiency of iodine would do which of the following?
 - a) Directly affect the synthesis of Thyroidin
 - b) Result in increased secretion of Thyroid stimulating hormone (TSH)
 - c) Result in decreased production of Thyrotropin-releasing hormone
 - d) result in increased basal metabolic rate
7. Which of the following acts to increase the release of Ca^{++} from the endoplasmic reticulum?
 - a) Diacylglycerol (DAG)
 - b) Inositol tri-phosphate
 - c) Parathyroid hormone
 - d) Calcitonin
8. Which of the following stimulates the synthesis of milk protein?
 - a) Luteinizing hormone (LH)
 - b) Prolactin (PRL)
 - c) Thyroid stimulating hormone (TSH)
 - d) Growth hormone (GH)
9. The release of which of the following is inhibited by Thyroxin?
 - a) LH
 - b) PRL
 - c) TSH
 - d) GH
10. Which of the following hormones promotes hypoglycemia?
 - a) Epinephrine
 - b) Norepinephrine
 - c) Insulin
 - d) Growth hormone
11. Increased reabsorption of water from the kidney is the major consequence of which of the following hormones?
 - a) Cortisol
 - b) Insulin
 - c) Vasopressin
 - d) Glucagon

12. Lack of glucocorticoids and mineralocorticoids might be a consequence of which of the following defects in the adrenal cortex?
- C 21-hydroxylase deficiency
 - testosterone deficiency
 - 17- α hydroxy progesterone deficiency
 - Androstenedione deficiency
13. Which of the following hormones can cause hypoglycemia without known effects on or gluconeogenesis?
- Thyroxin
 - Epinephrine
 - Glucocorticoids
 - Glucogen
14. Which of the following statements best describes Insulin?
- Its is antagonistic to that of glucagon
 - It is small peptide composed of a single chain bridged by disulfide groups
 - in does nor have a prohormone from
 - It promotes glucose absorption from the intestine and renal tubular cells
15. Which of the following is noted in Cushing's syndrome, a disease of the adrenal cortex?
- Decreased production of epinephrine
 - Excessive production of vasopressin
 - Decreased production of cortisol
 - Excessive production of cortisol
16. Which of the following statements best describes the mechanism of action of sex hormones?
- They bind specific membrane reseptors
 - They interact with DNA directly
 - They cause the release of the second messenger from the cell vevbrane
 - They enhance trancription when bound to receptors
17. Which of the flowing stimulates the production of progesterone by the corpus luteum?
- LH
 - PRL
 - TSH
 - GH
18. Which of the following pituitary hormones will if communcation from the hypothalamus is severed?
- GH
 - TSH
 - PRL
 - Vasopressin
19. Which of the following hormones is associated with diuresis?
- Oxytocin
 - Vasopressin
 - Prolactin
 - None of the above
20. A 73-year-old woman is transferred to an intensive care unit after she is found to be in septic shock. It is believed that she originally had a urinary tract infection and the bacteria seeded ger bloodstream. The critical care fellow is concerned she may not have an adeguete stress response in this situation. He orders a cosyntropin test, which evaluates the bodys ability to produce which of the following hormones?
- Oxytocin
 - Vasopressin
 - Cortisol
 - Corticotropin-releasing hormone
21. Which of the following hormones is secreted in response to Angiotensin II?
- Cortisol
 - Aldosterone
 - Thyroxin
 - Estrogen
22. A 73-year-old woman has been referred from an endocrilogist for weight gain, especially around the waist. She also has striae over the abdomen and rounded appearance to her face. She is found to nave CushiHg disease. Which of the following is found in this patient?
- Deceased absorption of glucose from the intestine
 - decreased lipolysis
 - Increased protein synthesis
 - Decreased liver glycogen stores
- No correct option, b) is the next ext correct option [I searched on multiple platform] It will be better if you search yourself.*
23. Hormones secreted by the posterior pituitary gland are made in the ____.
- Hypothalamus
 - Medulla oblongata
 - Cerebellum
 - Thalamus
24. What hormone promotes water retention by the kidneys?
- Follicle-stimulating hormone (FSH)
 - Prolactin
 - Antidiuretic hormone (ADH)
 - Melatonin

25. The primary chemical precursor for all steroid hormones is:

- a) Protein
- b) Carbohydrate
- c) Cholesterol
- d) Amino acids

26. A lack of iodine in the diet can lead to which condition?

- a) Diabetes mellitus
- b) Addison's disease
- c) Goiter (enlargement of the thyroid gland)
- d) Cushing's syndrome

27. The half life of insulin is

- a) < 3–5 minutes
- b) < 8–10 minutes
- c) < 15 minutes
- d) < 15 minutes

28. Following is a normal overnight fast and a cup of black coffee, a diabetic woman feels slightly nauseous and decides to skip breakfast. However she does take her shot of insulin. This may result in

- a) Heightened glycogenolysis
- b) Hypoglycemia
- c) Increased lipolysis
- d) Glycosuria

29. A hormone produced by corpus luteum and placenta, concerned with relaxation of pelvis tissue is

- a) HCG
- b) Chorionic somatomammotropin
- c) Relaxin
- d) Progestins

30. Melanocyte stimulating hormone is secreted by

- a) Pineal gland
- b) Anterior lobe of pituitary gland
- c) Posterior lobe of pituitary gland
- d) Intermediate lobe of pituitary gland

MCQ: Nucleic acids Metabolism

1. Rate controlling step of pyrimidine biosynthesis is catalyzed by:

- a) Orotidylate decarboxylase
- b) Aspartate transcarbamoylase
- c) Carbamoyl phosphate synthase II
- d) Orotate phosphoribosyl transferase

2. Which statement for purine biosynthesis is incorrect?

- a) Requires vitamin B₁₂
- b) Assembled on ribose phosphate
- c) Requires PRPP
- d) Requires glycine

3. Purines and pyrimidines are:

- a) Dietary essential
- b) Dietary non-essential
- c) Derived from essential fatty acids
- d) Derivatives of essential amino acid

4. The two nitrogens of the pyrimidine ring are contributed by:

- a) Glutamate and ammonia
- b) Glutamate and aspartate
- c) Glutamate and glutamine
- d) Aspartate and carbamoyl phosphate

5. Nitrogen or carbon atoms are contributed to the structure of the purine ring by amino acids, except:

- a) Glycine
- b) Glutamine
- c) Aspartate
- d) Glutamate

6. Lesch-Nyhan syndrome may lead to:

- a) Self-destructive behavior
- b) Gout
- c) Elevated levels of PRPP
- d) All of the above

7. Hyperuricemia can result from defect in enzymes, except:

- a) Carbamoyl phosphate synthase II
- b) HGPRTase
- c) PRPP synthase
- d) Glucose-6-phosphatase

8. The end product of purine metabolism in human is:

- a) Creatinine
- b) Uric acid
- c) Urea
- d) Ammonia

9. The salvage pathway for purines involves enzyme:
 - a) PRPP amidotransferase
 - c) HGPRTase**
 - b) PRPP synthase
 - d) Xanthine oxidase
10. Enzyme involved with immunodeficiency disease of purine metabolism is:
 - a) Adenosine deaminase**
 - b) Xanthine oxidase
 - c) PRPP synthetase
 - d) HGPRTase
11. Purine salvage occurs in the tissues, except:
 - a) RBC
 - c) Liver**
 - b) Brain
 - d) Polymorphonuclear leukocytes
12. Allopurinol is used in the treatment of:
 - a) Rickets
 - b) Cancer
 - c) Gout**
 - d) Pellagra
13. Lesch-Nyhan syndrome is due to the lack of:
 - a) HGPRTase**
 - b) APRTase
 - c) Adenosine deaminase
 - d) PRPP amidotransferase
14. Hereditary orotic aciduria type II is due to deficiency of the following enzyme:
 - a) Dihydro-orotate dehydrogenase
 - b) Orotate phosphoribosyl transferase
 - c) Aspartate transcarbamoylase
 - d) Orotidylate decarboxylase**

Replication, Transcription and Translation

1. A promoter site on DNA:
 - a) Transcribes repressor
 - c) Initiates transcription**
 - b) Codes for RNA polymerase
 - d) Regulates termination
2. A codon AUG is a:
 - a) Chain initiating codon**
 - b) Chain terminating codon
 - c) Releasing factor for peptide chains
 - d) Recognition site on the tRNA
3. Release of a polypeptide chain from a ribosome is catalyzed by:
 - a) Release factors**
 - b) Dissociation of ribosomes
 - c) Peptidyl transferase
 - d) Stop codons
4. In contrast to eukaryotic mRNA, prokaryotic mRNA:
 - a) Has a poly A tail**
 - b) Has 7-methylguanosine at the 5' end
 - c) Can be polycistronic**
 - d) None of the above
5. Erythromycin prevents synthesis of polypeptide:
 - a) By inhibiting binding of aminoacyl tRNA to mRNA
 - b) By inhibiting translocation reaction**
 - c) By inhibiting binding of tRNA to mRNA
 - d) Blocking mRNA formation from DNA
6. Formation of Okazaki fragments occur in the process of:
 - a) Transcription
 - c) Replication**
 - b) Translation
 - d) Reverse transcription
7. Termination of protein biosynthesis requires presence of:
 - a) Termination sequences "ter"
 - c) Nonsense codons**
 - b) Rho factor
 - d) Sigma factor
8. Which of the following enzyme joins Okazaki fragments?
 - a) DNA polymerase
 - b) DNA ligase**
 - c) RNA polymerase
 - d) Peptidyl transferase
9. Which the following enzyme fill the gap between Okazaki fragments?
 - a) DNA polymerase**
 - c) Translocase
 - b) RNA polymerase
 - d) Helicase

10. Which of the following nucleotide base is not present in codons?
 a) Adenine
 b) Guanine
 c) Thymine
 d) Cytosine
11. Total number of codons are:
 a) 64
 b) 61
 c) 62
 d) 63
12. Which of the following eukaryotic DNA polymerases is required for mitochondrial DNA synthesis?
 a) DNA polymerase α
 b) DNA polymerase β
 c) DNA polymerase γ
 d) DNA polymerase δ
13. Degeneracy of the genetic code means that:
 a) Codons are not ambiguous
 b) A given amino acid can be coded for by more than one base triplet
 c) A given codon can code for more than one amino acid
 d) There is no punctuation in the code sequence
14. Which of the following inhibits eukaryotic protein synthesis?
 a) Tetracycline
 b) Erythromycin
 c) Streptomycin
 d) Puromycin
15. Consensus sequence is:
 a) Initiation site of replication in eukaryotes
 b) Initiation site of replication in prokaryotes
 c) Initiation site of transcription in eukaryotes
 d) Initiation site of transcription in prokaryotes

Regulation of Gene Expression and Mutation

1. Restriction endonuclease has the following characteristics, except:
 a) Cut DNA in a sequence specific manner
 b) Named according to the bacteria from which they are isolated
 c) Most of the DNA sequences recognized by it are palindromic
 d) Cut DNA randomly
2. Most commonly used cloning vectors are:
 a) Bacterial plasmids
 b) Bacteriophages
 c) Cosmids
 d) All of the above
3. Cloning is a technique developed for:
 a) Amplifying the quantity of DNA in vivo
 b) Amplifying the quantity of DNA in vitro
 c) To search specific gene from DNA library
 d) To visualize specific DNA, RNA or proteins
4. Southern blot is a technique used for visualization of specific:
 a) RNA
 b) DNA
 c) Protein
 d) Carbohydrate
5. Polymerase chain reaction (PCR) is a technique for:
 a) Amplification of DNA in vitro
 b) Amplification of DNA in vivo
 c) Amplification of protein
 d) Amplification of RNA
6. Which one of the following enzymes is required for PCR?
 a) DNA ligase
 b) Thymidine kinase
 c) Taq polymerase
 d) Restriction endonuclease
7. Restriction endonucleases are enzymes:
 a) Used to joining DNA to cloning vector
 b) Which cleaves DNA at specific sequence
 c) Which cleaves DNA randomly
 d) Which digest DNA molecule from ends

Total 400 questions